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Variability of menstrual cycles by age, polycystic ovary syndrome, and early-life cycle irregularity in the Apple Women's Health Study

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Abstract

Objectives: Polycystic ovary syndrome (PCOS) is a common endocrine disorder, characterized by oligomenorrhea and androgen excess. Only a few studies have addressed the natural history of menstrual cycles among women with PCOS and/or irregular cycles, most with limited sample size and homogenous populations. We describe age-related differences in menstrual cycle length (MCL) and regularity for those with and without a diagnosis of PCOS.

Study Design: We included participants in the Apple Women's Health Study, a digital US cohort, who consented and enrolled between 11/2019 and 3/2024 and provided menstrual logging data (on days of menstrual bleeding) for at least three cycles without hormone use, pregnancy, or lactation. We evaluated MCL among three mutually exclusive groups: those reporting a diagnosis of PCOS (**group 1: PCOS**), those without PCOS but with early-life irregular (unpredictable) cycles (**group 2: early-life irregular**); and those without PCOS and with early-life regular (predictable) cycles (**group 3: early-life regular**). PCOS status was based on self-reported physician diagnosis. "Early-life irregular" was defined as reporting not spontaneously establishing regular cycles within 4 years after menarche for those who did not report a PCOS diagnosis. Menstrual cycles were categorized by participant's age into eight groups (<20, 20–24, 25–29, 30–34, 35–39, 40–44, 45–49, 50+ years). We used linear mixed effect (LME) models with random participant-specific intercepts to estimate a **mean MCL** by age groups. Log-linear models for residual variance were used in the LME models to estimate how within-individual standard deviations (SDs) of MCL (**as a measure of cycle irregularity**) vary across age groups. P-values for interactions were calculated to compare across groups.

Results: Data from 160,206 menstrual cycles across 15,586 participants were analyzed, including 18,875 cycles from 1,842 participants with PCOS. Mean age at enrollment was 33 years for all groups. Prevalence of obesity (BMI ≥ 30.0) was 62% in the PCOS group, 36% in the early-life irregular group, and 35% in those with early-life regular cycles. Around 22% of the cycles in the PCOS group were 30–34 years of age and 24% were ages 35–39. MCL patterns across the reproductive lifespan differed among groups. Compared to group 3 with early-life regular cycles, group 1 (PCOS) and group 2 (early-life irregular) consistently had longer mean MCL in early reproductive years, with the difference between groups decreasing with age up to ≤ 40 years. Covariate-adjusted results showed similar trends. Decreases in cycle irregularity (within-individual SD) for groups 1 and 2 were also seen across the 20s and 30s age groups.

Conclusions: At younger ages, persons with PCOS or early-life irregularity had longer and more irregular cycles than those with regular cycles, as expected. These differences among groups diminished with age, as cycle characteristics of those with PCOS or early-life irregularity became more similar to those of persons with early regular cycles.

Tweetable statement :

A large cohort found age-related variations in menstrual cycle length and irregularity for people with PCOS or irregular cycles, offering new insights into menstrual health over the reproductive life course.

Keywords

polycystic ovary syndrome; cycle length; lifecourse; menstrual regularity

Introduction

In mid-reproductive years, menstrual cycles typically range from 25–30 days with 28 days being most common.^{1–5} Cycles documented through paper-based prospective records or interview tend to become shorter with age and often become irregular before menstrual cessation at menopause.⁶ Chiazze et al. observed highest cycle variability among 2,316 US/Canadian women aged <25 years, with cycle length declining to minimum at ages 35–39 years,⁷ consistent with Vollman's findings among 592–656 healthy Swiss women.^{8,9} Treloar et al. similarly confirmed patterns of decreasing cycle length by both chronological and gynecological age among over 2,700 females in Minnesota, US.⁶ Vollman also identified decreasing cycle length with increasing years since menarche among adolescent girls followed for 12 years.¹⁰ A recent study by Bull et al. using cycles collected from a mobile app identified a decrease in mean cycle length overall among ages 25–45.¹ These foundational studies on menstrual cycle variations^{6–8,10,11} did not differentiate how these age-related patterns vary by underlying ovulatory disorders or early-life menstrual characteristics.

Polycystic ovary syndrome (PCOS) is a common ovulation disorder, characterized by abnormally long or irregular cycles, and androgen excess.^{12,13} Individuals with PCOS may exhibit irregular cycles from adolescence¹⁴ through adulthood. While limited studies with small sample sizes suggest cycle length attenuation with age, the extent and timing across the reproductive lifespan remains uncertain. Jacewicz-Swiecka et al. conducted a longitudinal study of 31 Polish patients diagnosed with PCOS during 2003–2009 and reassessed during 2015–2017 (median age: 35 years).¹⁵ They found oligomenorrhea decreased from 98% at baseline to 42% during follow-up (median: 10 years). Elting et al. studied 346 participants in the Dutch Aging in Polycystic Ovarian Syndrome cohort and reported a negative correlation between age and cycle length, which remained statistically significant after adjusting for BMI.¹⁶ Elting et al. subsequently examined 27 patients with PCOS and found that those who achieved regular cycles with increased age (median age: 40 years) had a lower ovarian follicle count than those with continued oligomenorrhea/amenorrhea, suggesting a potential mechanism for regained regularity.¹⁷ However, these observations from small, clinically-referred cohorts potentially limit generalizability.

Similarly, little is known about whether comparable patterns in cycle length and regularity may be observed among individuals without a reported PCOS diagnosis but who exhibit menstrual irregularities during the early period of the reproductive lifespan. The POMP study prospectively evaluated Dutch adolescents and their menstrual cycle characteristics, and found an association between oligomenorrhea and serum laboratory values consistent with PCOS. This suggests that a failure to establish regular menstrual cycles in early reproductive life is a risk factor for future diagnosis of PCOS.^{18,19}

The primary aim of this study was to assess cycle length and irregularity across age groups among three groups within a large US cohort: those with a self-reported diagnosis of PCOS, those without PCOS but with early-life irregular cycles, and those without PCOS and reporting early-life regular cycles.

Materials and Methods

Study population

The Apple Women's Health Study is a prospective digital cohort study in the United States (US). Users of the Apple Research app on their iPhone were eligible to participate if they had ever menstruated at least once in life, live in the US, were at least 18 years old (19 in Alabama/Nebraska; 21 in Puerto Rico), and were able to communicate in English. Eligibility also required sole use of an iCloud account and an iPhone. Enrollment began on November 14, 2019, and is ongoing. Participants provided written informed consent at enrollment. This study was approved by the Institutional Review Board at Advarra. Details have been described previously.²⁰ Demographic, medical history, and reproductive surveys were completed upon enrollment. We included participants who consented and enrolled between 11/2019–3/2024, provided data of ≥ 3 logged cycles (not necessarily consecutive, although 80% of participants had ≥ 3 consecutive cycles) that have been without hormone use (any exogenous hormone for any reason, i.e., including only unmedicated cycles), pregnancy, or lactation in the past month. Participants in this analytical dataset were also required to have responded to survey questions regarding self-reported PCOS status and time from menarche to establishing regular cycles. A detailed participant flowchart is shown in Figure S1, and a conceptual model in Figure S2. The final study population included 15,586 participants with 160,206 cycles. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guideline, with the checklist available in supplemental materials.²¹

Logged cycle length

Participants are able to log their menstrual flow days using the Cycle Tracking feature with the Apple Health app or other third-party apps that the participant allows to write to the Health app.²² Menstrual cycle length (MCL) for each cycle was calculated by subtracting the reported date of first bleeding from the subsequent reported first bleed date using previously established definitions and algorithms.²² Cycles $<10/>90$ days were excluded as potentially inaccurate based on our previous work.²² As detailed above, we only included cycles confirmed without hormone use/pregnancy/lactation. A sensitivity analysis removing individuals who reported pregnancy/lactation within 2 years before enrollment (5–7%) was conducted to account for possible post-delivery hormonal changes and breastfeeding-related ovulation suppression.

Mutually exclusive groupings

We evaluated cycle characteristics among the following mutually exclusive groups: **Group 1** (PCOS) comprised those who reported having been diagnosed with PCOS; **Group 2** (early-life irregular) were those without PCOS but reported not developing regular cycles within 4 years after menarche; and **Group 3** (early-life regular) were those without PCOS who spontaneously established regular cycles within 4 years of menarche. **Figure 1 describes these groups.** Age at each cycle was calculated (cycle year – birth year), and grouped into intervals ($<20/20\text{--}24/25\text{--}29/30\text{--}34/35\text{--}39/40\text{--}44/45\text{--}49/50+$ years).

Covariates

Other covariates from participants' baseline survey response included: (1) *variables for stratification*: sociodemographic [race/ethnicity, subjective socioeconomic status (SES) scale],²³ BMI, and relevant conditions [endometriosis, fibroids, infertility, uterine/cervical polyps, premenstrual syndrome or premenstrual dysphoric disorder (PMS/PMDD), hyperprolactinemia, hyper- or hypothyroidism, or diabetes/prediabetes] and (2) *covariates for adjustment in models*, including physical activity (exercise minutes/week), sleep (ever experiencing sleep difficulty), stress [derived Perceived Stress Scale 4 (PSS-4) score],²⁴ diet variables (any low calorie/carb/fat diet, any high fat diet, any high protein diet, or any vegetarian/vegan diet), smoking (current/past/never), alcohol frequency, e-cigarette or marijuana use (current/past/never). Covariates details are summarized in Table 2.

Statistical approaches

Findings were stratified by grouping (Figure 1). Initially, unadjusted cubic B-spline models with 4 degrees of freedom were used to assess the overall MCL pattern by age in years, using all eligible cycles. Then, for the recorded cycles contributed by each participant, linear mixed effect (LME) models with random participant-specific intercepts were used to estimate the **mean MCL by age** groups of logged cycles. Linear age trends from 18 to 44 years were evaluated within each group, with age as a continuous variable (p-for-trends for ages 18–44). Age and PCOS/irregular cycle grouping interactions were evaluated by including an interaction term in the LME models in the full analytical dataset. Additionally, log-linear models for residual variance were used to estimate how **within-individual standard deviation (SDs) of MCL (as a measure of cycle irregularity)** varied across age groups. All LME-related analyses included an unadjusted model (model 1, not adjusting for any covariates), and a model adjusted for all baseline covariates (model 2).

As secondary analyses, we stratified models by race/ethnicity, SES, BMI, and other reported medical conditions. Additionally, we evaluated a subgroup with potentially more severe and persistent irregularities: participants who reported having PCOS and with early-life irregular cycles (n=928). Among individuals who reported age at menarche, we calculated gynecological age of each logged cycle as years since menarche, and estimated covariate-adjusted MCL and cycle variability patterns by gynecological age (<10/10–14/15–19/20–24/25–29/30–34/35–39/40+ years). Lastly, we evaluated a subset of participants who shared their medication records and summarized MCL for selected medications that may impact menstrual cycles.²⁵

Analyses were conducted in Python, version 3.6 (Python Software Foundation) and R, version 4.1.2 (R Project for Statistical Computing). All statistical tests were two-sided with 95% CIs. P-value < .05 was considered statistically significant.

Results

Table 1 shows the age distributions across 160,206 cycles from 15,586 participants. Among the 18,875 cycles of participants with PCOS (group 1), the most common age at report was 30–39 years (46%), with similar age distributions in groups 2 (early-life irregular) and 3

(early-life regular). Baseline characteristics are described in Table 2, where 27% of group 1 and 30% of group 2 had low SES, compared to 23% of group 3. For BMI, 62% of group 1 reported BMI ≥ 30.0 compared to 36% and 35% in groups 2 and 3. Most were nulliparous (66% of group 1, 59% of group 3). Compared to group 3, group 1 engaged in less physical activity (20% reporting >150 minutes/week vs. 26%), more low carb diet (17% vs. 9%), and reported more sleep difficulties (51% vs. 34%). Groups 1, 2, and 3 had similar numbers of recorded cycles per person (means: 10.9, 10.1, and 10.3 cycles, respectively; Figure S1). When evaluating self-reported age at diagnosis data (Table S1), most participants (74%) in group 1 had PCOS diagnosed in or after 2006 (the year when the AE-PCOS 2006 criteria was released to the public),^{26,27} though information on the specific diagnostic criteria applied at each diagnosis remains lacking.

Figure 2 shows the overall unadjusted MCL trends from spline models, where MCL decreased with age in all groups, with varying ages of shortest MCL before increasing again around typical ages of perimenopause. Figure 3 and Table 3 shows the estimated mean (95% CI) of MCL by age group from LME models. In these results, we mainly describe the data at ages 18–44; after that, the potential influence of perimenopause may impact interpretations. In unadjusted models, group 1 or 2 had longer mean MCLs than group 3. As age increased, mean MCL in group 1 decreased, reaching 31.5 days (95% CI 30.8–32.2) at age 40–44. In comparison, mean MCL in group 3 decreased to 28.4 days (95% CI 28.2–28.6) at age 40–44. Covariate-adjusted models showed similar patterns, with the MCL decrease across ages 18–44 years being greatest in group 1 (age*group interaction $p<0.001$).

Figure 4 and Table 4 provide estimated cycle length variability (95% CIs) by age. Among those aged 18–44, cycle variability decreased with age in group 1, while remaining stable in group 3 until age category 40–44. Covariate-adjusted models maintained similar patterns.

Subgroup analyses (Figures S3–S7) suggest little evidence of effect modification (p -for-interactions >0.05) by race/ethnicity, SES, BMI, parity or self-reported conditions, though the magnitude of differences in MCL were smaller in certain strata. Among a subset of participants with PCOS and early-life irregular cycles ($n=928$, Tables S2–S3), cycle length and variability were higher than in the three main groups (group 1, 2, and 3), while still decreasing with age groups (Tables S4–S5, Figure S8). Sensitivity analyses excluding recent pregnancy/lactation confirmed main findings (Tables S6–S7). When using gynecological age categories, cycle length and variability patterns (Supplemental Figure S9) were similar to the main findings based on chronological age; all individuals (with or without PCOS) tended to reach similar cycle length and variability at 35–39 years of gynecological age. Medication user distributions and their cycle patterns are summarized in Tables S8–S9.

Comment

Principal findings

In this analysis utilizing logged cycle data from a large, US-based digital cohort, cycle length and variability decreased linearly with advancing age (18–44 years) among those with PCOS and/or early-life irregular cycles. Despite these individuals initially having longer cycle length and larger variability than females with early-life regular cycles, those

with PCOS or early-life irregular cycles had more similarity to the comparison group with increase in age. Cycle length and variability converged across all groups around ages 45–49. While these results are consistent after control of known confounders, they may be influenced by unmeasured factors including perimenopause status, which is worthy of further investigation.

Results in the context of what is known

Our findings align with the limited historical data on MCL patterns over age among people with PCOS. Our study substantially expands on previous small studies (sample sizes 31 to 346) by providing more generalizable results. We found that mean cycle length and presence of irregularity decrease with age among those with PCOS (before typical age of menopausal transition), approaching those of women without PCOS by their late reproductive years (6–8). Unlike previous studies that primarily recruited patients referred for care from specific geographical regions/clinics, our study is the first to comprehensively evaluate cycle length and variability across the reproductive lifespan (age 18 to perimenopausal age) for those with PCOS, using logged cycle data from >15K participants in a large US cohort with diverse sociodemographic and health characteristics.

As age increases and the follicular pool becomes depleted, less inhibin is released from relatively less follicles, leading to reduced negative feedback on FSH, resulting in elevated FSH levels. This leads to earlier recruitment of a dominant follicle, and a shorter follicular phase of the cycle, shortening MCL. This process occurs in persons with PCOS too, leading to shorter MCL in the perimenopause. This is a reason for caution in interpreting the MCL changes that occur in older participants in this cohort, and so future research on older patients with PCOS will be required to further characterize this change.¹⁶

Our study also evaluated cycle patterns among individuals without diagnosed PCOS but who reported cycle irregularity during adolescence, a group that has not been previously examined. It is possible that some in this group (group 2) could have met the criteria for PCOS but have not yet received a diagnosis of PCOS. Approximately 30% women report 2 years or more between onset of symptoms and diagnosis of PCOS, with many requiring seeing multiple health providers prior to diagnosis.²⁸ Table S1 summarizes the time periods when participants with PCOS were diagnosed, and the diagnostic criteria in use at the time. Variations in criteria over time may also contribute to likelihood of diagnosis. Other potential causes of oligo/anovulation may also lead to self-report of irregularity without PCOS diagnosis (group 2). Despite the heterogeneity of group 2, this group is useful to describe as they are at risk of long-term health outcomes associated with persistent irregular cycles.^{14,29}

Clinical implications

This study provides evidence of dynamic changes in menstrual cycle length and variability with increase in age among participants with irregular cycles (with or without PCOS). Clinicians can support counseling regarding age-related establishment of regular cycles, and educate patients about expected changes across the life stages.³⁰ Expected changes in cycle length and variability in populations with PCOS approaching perimenopause are consistent

with previous findings, while our findings among individuals without a PCOS diagnosis but with early-life irregular cycles may be useful for inclusive counseling of patients with irregular cycles across varying life stages.^{31–33}

Research implications

Digitally-collected longitudinal data on menstrual cycle characteristics over the life course can provide valuable information on the life course changes in menstruation and its health implications. Utilizing multiple cycles per individual in a large, digital cohort setting allowed for robust estimation of cycle length and variability across age groups from early adulthood through menopause. Future studies are needed to describe within-person cycle length trajectories. Our findings also carry health implications. In the Nurses' Health Study II, among 75,546 premenopausal participants who were followed over 24 years,³⁴ cycle irregularity during adolescence/adulthood were associated with higher risk of type 2 diabetes; specifically, the magnitude of associations with type 2 diabetes were stronger among those who reported irregular cycles at later life stages (e.g., age 29–46 vs. age 14–22 years). Individuals with irregular cycles at age 29–46 also had the largest hazard ratio for cardiovascular diseases.³⁵ These results suggest persistent indications of oligomenorrhea may predict increased risks of adverse outcomes. Our study adds to our understanding of the natural progression of irregular cycles among those with PCOS, information that is potentially useful for categorizing individuals with PCOS and/or early-life irregular cycles into risk categories by age and symptom persistence. Further prospective validations will help with this risk stratification.

Strengths and limitations

Our large sample size is a major strength, providing sufficient statistical power to detect cycle differences and evaluate modifications by health characteristics. Cycle data collection through available apps allows more consistent and accurate evaluation of cycle characteristics than retrospective reporting.²²

Limitations include self-reporting of clinician-diagnosed PCOS status. However, group 2 may include some with PCOS who have not been diagnosed, especially in the younger age groups, given that diagnosis is often delayed.^{36,37} Accurate logging of cycle data is not always guaranteed even in prospective studies, but we excluded cycles <10 days or >90 days in length to reduce the most extreme errors. Currently insufficient data on age at menopause/perimenopause initiation prevented further detailed categorization by perimenopausal status. Also, possible residual error from intermenstrual spotting/bleeding may be present despite our use of previously established approaches/algorithms, potentially causing differential misclassification of cycle length amongst participants with PCOS/irregular cycles compared with those with regular cycles. Severe PCOS cases treated with oral contraceptives are not included in this analysis. While we included cycles confirmed without hormone use in the past month, residual hormone effect may be present for some. We did not have sufficient data to exclude medications use such as metformin, GLP-1, or spironolactone used concurrently with logged cycles, which may influence cycle length. However, based on our exploration, participants who were ever users tended to have cycles lengths closer to 28 days, suggesting that inclusion of these individuals may result in underestimation of

cycle length and variability in the PCOS group. Thus, generalizability to all individuals with PCOS, especially severe cases, may be limited, though underlying MCL attenuation biology is likely similar. Finally, while smartphones are widely used by young people in the US, the generalizability of this study is limited by the requirement that participants have iPhones.

Conclusions

This study provides the most comprehensive description to date of menstrual patterns across the life course for females with PCOS or long-term oligomenorrhea. The findings are useful to these persons and their health care providers as an aid in understanding the natural course of their condition, and in managing the many aspects of life, such as family planning, that are related to menstrual function.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Sharing Statement

Aggregated data that support the findings of this study may be available upon reasonable request from the corresponding author & senior author. Any request for data will be evaluated and responded to in a manner consistent with legal obligation and policies intended to protect participant confidentiality, language in the Study protocol, and informed consent form.

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AJOG at a Glance:**Why was this study conducted?**

Limited research exists on menstrual cycle patterns across reproductive life stages for individuals with PCOS or early-life irregular cycles.

What are the key findings?

We analyzed >160K logged cycles from >15K participants, observing distinct patterns of menstrual cycle length (MCL) and variability across age groups:

- Persons with PCOS or early-life irregular cycles had longer MCL and greater cycle irregularity compared to those whose early-life cycles were regular
- During ages 18–44, MCL and variability decreased with age in all groups, most strongly among those with PCOS/early-life irregularity.

What does this study add to what is already known?

- This unique large-scale digital study of age-related menstrual cycle variations provides better understanding of cycle dynamics among persons affected by PCOS or irregular cycles.

Sample size distribution by three mutually exclusive group

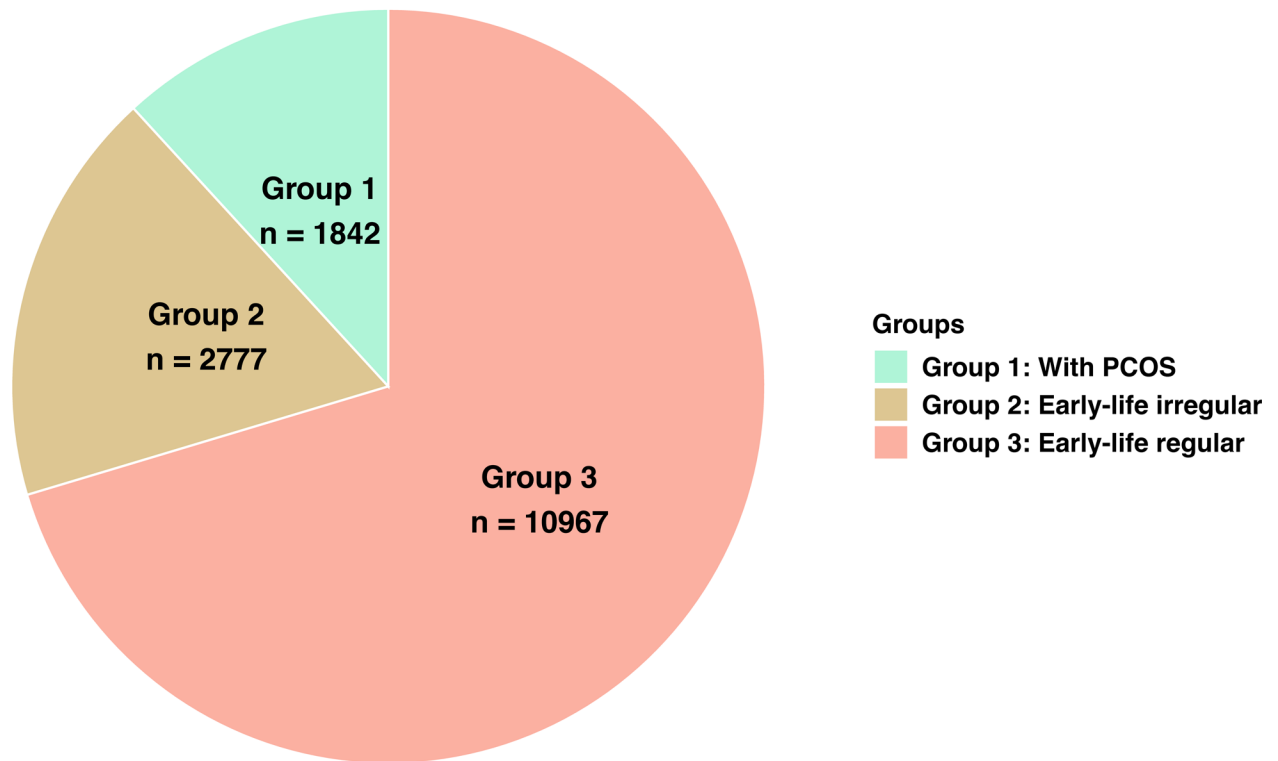


Figure 1.

Definitions for three mutually exclusive groups representing varying states of PCOS and earlier onset/persistently irregular cycles. Group 1: **PCOS** (all participants who reported PCOS).

Group 2: **Early-life irregular** (participants who did not report PCOS, but reported not spontaneously establishing regular menses within 4 years since menarche).

Group 3: **Early-life regular** (participants who did not report PCOS and reported being able to spontaneously establish regular menses within 4 years since menarche).

Group 3 represents the least severity and was used as the referent group in this analysis.

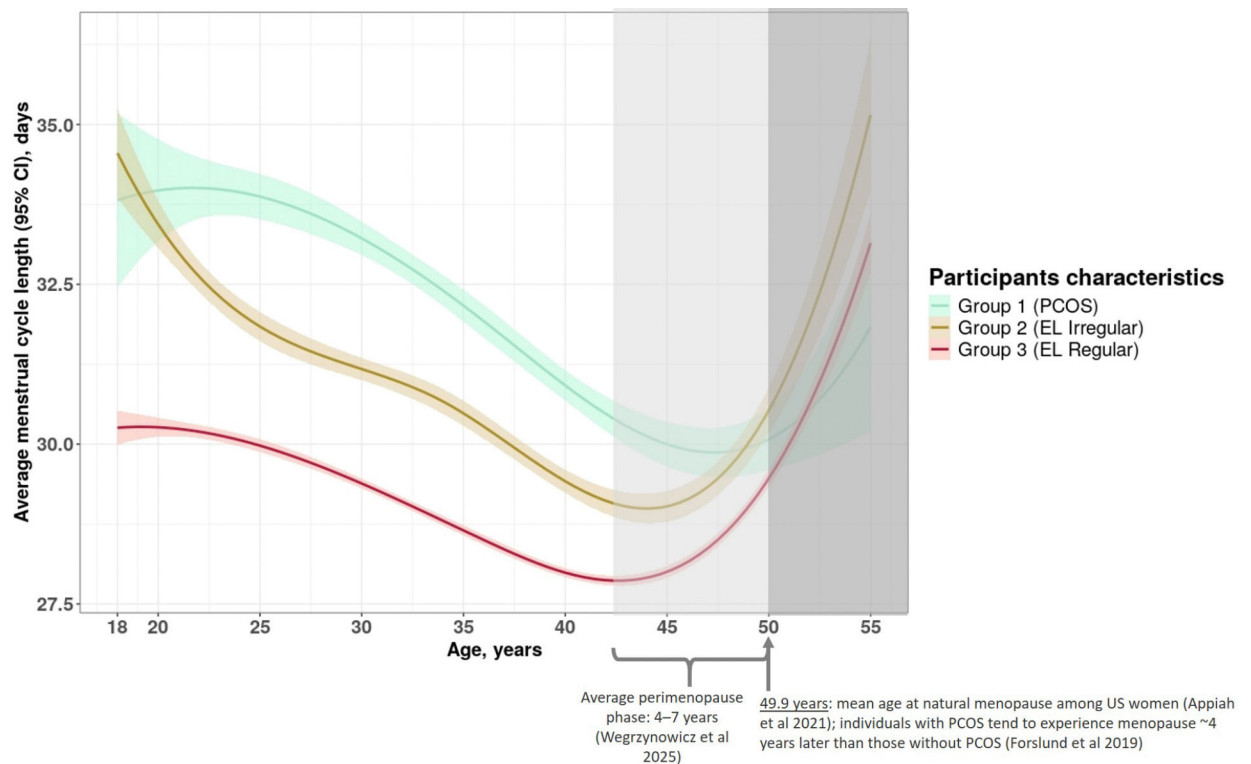


Figure 2.

Unadjusted mean menstrual cycle length (95% CI) by age in years from spline model of the three mutually exclusive groups.

Abbreviations: PCOS, polycystic ovary syndrome; EL, early-life.

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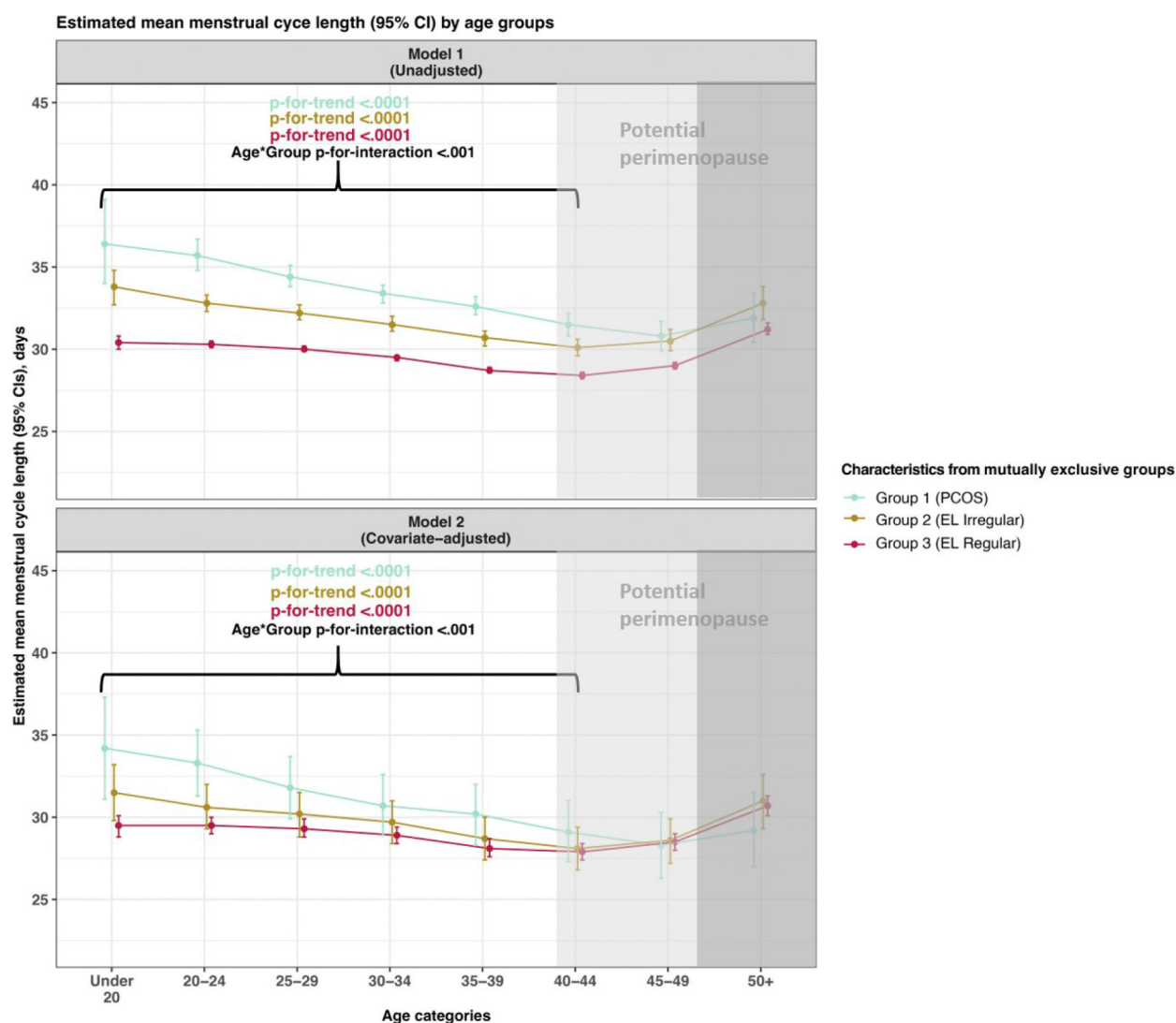


Figure 3.

Estimated mean menstrual cycle length (95% CI) by age group.

Abbreviations: PCOS, polycystic ovary syndrome; EL, early-life.

Model 1: Unadjusted model. Model 2: Adjusted for baseline covariates, including physical activity (exercise minutes per week), sleep (ever sleep difficulty), stress (PSS-4 score), diet (low calorie/carb/fat diet, high fat diet, high protein diet, vegetarian/vegan diet), smoking (current, past, never), alcohol (never, monthly or less, 2–4 times per month, 2–3 times per week, 4+ times per week), e-cigarette (current, past, never), or marijuana use (current, past, never).

Among age groups 18 to 44: P-for-trend estimated within each of the 3 mutually exclusive groups (group 1, 2, and 3 respectively) via including age (in years) as a continuous predictor in the linear mixed effect (LME) models with random participant-specific intercepts. P-for-interaction was evaluated for the 3 mutually exclusive groups via including an age*group interaction term (using age in years) in the linear mixed effect (LME) models with random participant-specific intercepts.

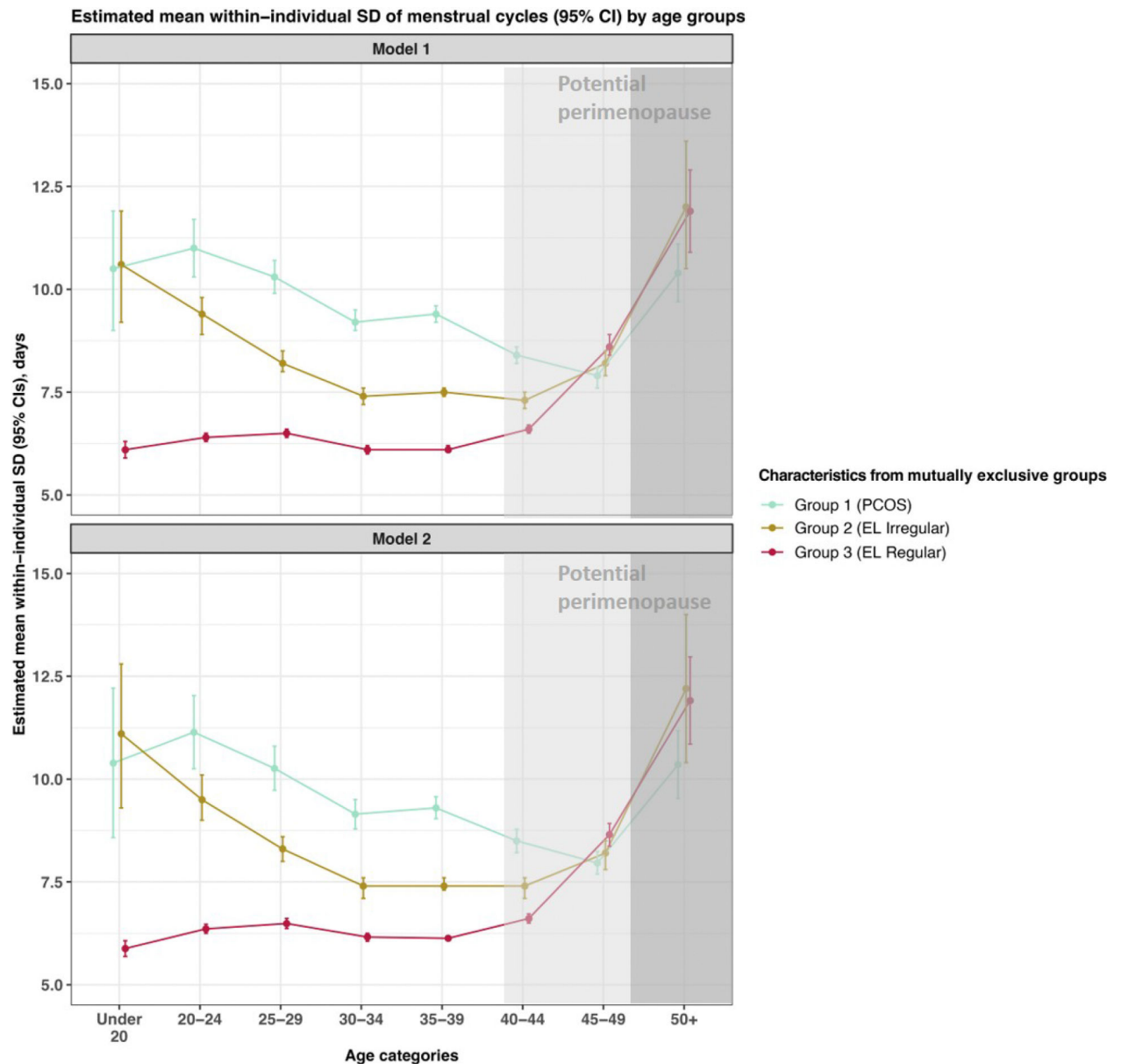


Figure 4.

Estimated cycle length variability (within-individual standard deviation) by age group.

Abbreviations: PCOS, polycystic ovary syndrome; EL, early-life.

Model 1: Unadjusted model. Model 2: Adjusted for baseline covariates, including physical activity (exercise minutes per week), sleep (ever sleep difficulty), stress (PSS-4 score), diet (low calorie/carb/fat diet, high fat diet, high protein diet, vegetarian/vegan diet), smoking (current, past, never), alcohol (never, monthly or less, 2–4 times per month, 2–3 times per week, 4+ times per week), e-cigarette (current, past, never), or marijuana use (current, past, never).

Table 1.

Age distributions of the 15,586 AWHs participants with data of 160,206 menstrual cycles, across the three mutually exclusive groups of PCOS and early-life regularity status.

Age distributions ^a	Group 1: With PCOS (18,875 cycles)	Group 2: Early life irregular (28,054 cycles)	Group 3: Early life regular (113,277 cycles)
Total N of cycles	18,875	28,054	113,277
Total N of participants	1,842	2,777	10,967
Age (chronological) at reported menstrual cycle, median (IQR)	36 (30 – 41)	34 (27 – 41)	36 (29 – 42)
Age (chronological) at menstrual cycle categories, n cycles (%):			
Under 20	118 (0.6)	591 (2.1)	2,709 (2.4)
20–24	1,254 (6.6)	3,756 (13.4)	13,054 (11.5)
25–29	2,962 (15.7)	4,828 (17.2)	15,578 (13.8)
30–34	4,125 (21.9)	5,630 (20.1)	20,520 (18.1)
35–39	4,458 (23.6)	4,969 (17.7)	21,806 (19.3)
40–44	3,400 (18.0)	4,312 (15.4)	20,582 (18.2)
45–49	1,839 (9.7)	3,006 (10.7)	14,295 (12.6)
50+	719 (3.8)	962 (3.4)	4,733 (4.2)
Gynecological age at reported menstrual cycle (among those with reported age at menarche), median (IQR)	23 (18 – 29)	21 (15 – 28)	24 (17 – 30)
Gynecological age at reported menstrual cycle (among those with reported age at menarche), n cycles (%):			
Under 5	535 (2.8) ^c	64 (0.2)	119 (0.1)
5–9		1862 (6.6)	7007 (6.2)
10–14	1891 (10.0)	4701 (16.8)	13220 (11.7)
15–19	3226 (17.6)	5107 (18.2)	16722 (14.8)
20–24	4382 (23.2)	5566 (19.8)	20889 (18.4)
25–29	4058 (21.5)	4515 (16.1)	21176 (18.7)
30–34	2773 (14.7)	3608 (12.9)	17845 (15.8)
35–39	1257 (6.7)	1953 (7.0)	9353 (8.3)
40–44	227 (1.2)	330 (1.2) ^b	1800 (1.6)
45–49	17 (0.7) ^b		91 (0.1)
50+			29 (0.0)

^aChronological age was evaluated as the main exposure variable in our analyses. Distributions of gynecological age (age at each logged cycle minus age at menarche) are provided in the Table for illustration purposes.

^bAggregated number due to too few cycles in the 45+ years gynecological age category.

^cAggregated number due to too few cycles in the <5 years gynecological age category (0.0%).

Table 2.
Baseline characteristics at enrollment among 15,586 participants of the three mutually exclusive groups.

Baseline characteristics at enrollment from self-report, n participants (%) ^a	Group 1: With PCOS (n = 1,842)	Group 2: Early life irregular (n = 2,777)	Group 3: early life regular (n = 10,967)
Age at enrollment, years:			
Median (IQR)	33 (28 – 39)	31 (25 – 38)	33 (26 – 40)
Under 20	31 (1.7)	133 (4.8)	536 (4.9)
20–24	182 (9.9)	481 (17.3)	1547 (14.1)
25–29	371 (20.1)	535 (19.3)	1781 (16.2)
30–34	471 (25.6)	595 (21.4)	2068 (18.9)
35–39	378 (20.5)	441 (15.9)	2020 (18.4)
40–44	245 (13.3)	338 (12.2)	1702 (15.5)
45–49	127 (6.9)	204 (7.3)	1006 (9.2)
50+	37 (2.0)	50 (1.8)	307 (2.8)
Race and ethnicity:			
Non-Hispanic white	1299 (70.5)	1941 (69.9)	7702 (70.2)
Non-Hispanic Black	81 (4.4)	141 (5.1)	671 (6.1)
Asian	58 (3.1)	93 (3.3)	495 (4.5)
Hispanic	153 (8.3)	230 (8.3)	774 (7.1)
Multiple races	213 (11.6)	295 (10.6)	1063 (9.7)
Other race(s)	35 (1.9)	70 (2.5)	229 (2.1)
Subjective socioeconomic status: ^b			
Low (scale 0–3)	505 (27.4)	824 (29.7)	2514 (22.9)
Medium (scale 4–5)	838 (45.5)	1210 (43.6)	4798 (43.7)
High (scale 6–9)	495 (26.9)	741 (26.7)	3641 (33.2)
Body mass index, kg/m²:			
Underweight (<18.5)	17 (0.9)	91 (3.3)	283 (2.6)
Healthy weight (18.5–24.9)	302 (16.4)	966 (34.8)	3856 (35.2)
Overweight (25.0–29.9)	360 (19.5)	689 (24.8)	2802 (25.5)
Obese (≥30.0)	1137 (61.7)	985 (35.5)	3811 (34.7)
Nulliparous	1222 (66.3)	1916 (69.0)	6421 (58.5)
Parity, mean ± SD	0.5 ± 0.9	0.6 ± 1.0	0.7 ± 1.2
Age at menarche			
Age 7 or younger	7 (0.4)	5 (0.2)	17 (0.2)
Age 8	20 (1.1)	16 (0.6)	63 (0.6)
Age 9	90 (4.9)	91 (3.3)	336 (3.1)
Age 10	185 (10.0)	188 (6.8)	782 (7.1)
Age 11	335 (18.2)	464 (16.7)	1962 (17.9)
Age 12	480 (26.1)	799 (28.8)	3095 (28.2)
Age 13	358 (19.4)	582 (21.0)	2257 (20.6)
Age 14	158 (8.6)	308 (11.1)	937 (8.5)
Age 15	73 (4.0)	145 (5.2)	369 (3.4)
Age 16 or older	78 (4.2)	142 (5.1)	209 (1.9)

Baseline characteristics at enrollment from self-report, n participants (%) ^a	Group 1: With PCOS (n = 1,842)	Group 2: Early life irregular (n = 2,777)	Group 3: early life regular (n = 10,967)
Gynecological age at baseline (among those with reported age at menarche), mean ± SD in years	21.5 ± 7.6	19.7 ± 8.3	21.6 ± 8.7
Gynecological age categories at baseline (among those with reported age at menarche):			
Gynecological age ≤3 years	0 (0.0)	4 (0.1)	5 (0.0)
Gynecological age >3 years	1784 (100.0)	2736 (99.9)	10022 (100.0)
Number of years since last hormone use (for past hormone users) at baseline, mean ± SD			
Mean ± SD	5.5 ± 6.0	5.6 ± 6.2	6.3 ± 6.7
Median (IQR)	3 (1 – 9)	3 (1 – 8)	4 (1 – 10)
Total number of years of past hormone use at baseline, mean ± SD			
Mean ± SD	6.8 ± 7.1	6.3 ± 7.1	5.0 ± 6.9
Median (IQR)	5 (0 – 12)	4 (0 – 11)	0 (0 – 9)
Self-reported endometriosis	181 (9.8)	145 (5.2)	371 (3.4)
Self-reported fibroids	171 (9.3)	108 (3.9)	535 (4.9)
Self-reported infertility	349 (18.9)	85 (3.1)	411 (3.7)
Self-reported PMS/PMDD	365 (19.8)	254 (9.1)	865 (7.9)
Self-reported hyperprolactinemia	46 (2.5)	24 (0.9)	68 (0.6)
Self-reported hyper- or hypothyroidism	349 (18.9)	232 (8.4)	1008 (9.2)
Self-reported prediabetes	397 (21.6)	142 (5.1)	477 (4.3)
Self-reported type 1 diabetes	13 (0.7)	17 (0.6)	42 (0.4)
Self-reported type 2 diabetes	134 (7.3)	42 (1.5)	166 (1.5)
Exercise minutes per week: ^c			
None	174 (9.4)	242 (8.7)	808 (7.4)
1–75	830 (45.1)	1059 (38.1)	3974 (36.2)
76–150	460 (25.0)	623 (22.4)	2750 (25.1)
151–300	250 (13.6)	449 (16.2)	1931 (17.6)
>300	113 (6.1)	280 (10.1)	970 (8.8)
Low calorie diet	150 (8.1)	168 (6.0)	706 (6.4)
Low carb diet	308 (16.7)	197 (7.1)	932 (8.5)
Low fat diet	69 (3.7)	77 (2.8)	302 (2.8)
High fat diet	26 (1.4)	30 (1.1)	125 (1.1)
High protein diet	180 (9.8)	166 (6.0)	695 (6.3)
Vegetarian diet	82 (4.5)	168 (6.0)	671 (6.1)
Vegan diet	16 (0.9)	70 (2.5)	244 (2.2)
Ever reported having trouble sleeping to health professionals	935 (50.8)	1145 (41.2)	3731 (34.0)
Perceived stress Scale (PSS-4) score: ^d			
Median (IQR)	7 (4 – 9)	7 (4 – 9)	6 (4 – 9)
Alcohol use and frequency in past year:			
Never	567 (30.8)	831 (29.9)	3394 (30.9)
Monthly or less	673 (36.5)	803 (28.9)	2902 (26.5)
Two to four times a month	327 (17.8)	513 (18.5)	2033 (18.5)

Baseline characteristics at enrollment from self-report, n participants (%) ^a	Group 1: With PCOS (n = 1,842)	Group 2: Early life irregular (n = 2,777)	Group 3: early life regular (n = 10,967)
Two to three times a week	149 (8.1)	298 (10.7)	1250 (11.4)
Four or more times a week	111 (6.0)	208 (7.5)	854 (7.8)
Tobacco use:			
Never smoked	1231 (66.8)	1822 (65.6)	7443 (67.9)
Past smoker	382 (20.7)	505 (18.2)	1930 (17.6)
Current smoker	182 (9.9)	262 (9.4)	876 (8.0)
E-cigarette use:			
Never used	1208 (65.6)	1728 (62.2)	7353 (67.0)
Past use	412 (22.4)	581 (20.9)	2024 (18.5)
Current use, some days	71 (3.9)	118 (4.2)	378 (3.4)
Current use, every day	134 (7.3)	213 (7.7)	650 (5.9)
Marijuana use:			
Never used	609 (33.1)	867 (31.2)	3665 (33.4)
Past use	723 (39.3)	1020 (36.7)	4149 (37.8)
Current use, some days	261 (14.2)	433 (15.6)	1513 (13.8)
Current use, every day	202 (11.0)	291 (10.5)	958 (8.7)
Reported pregnancy/breastfeeding in the last 2 years. N (%)	136 (7.4)	150 (5.4)	599 (5.5)

Abbreviations: PMS/PMDD, Premenstrual syndrome (PMS) or premenstrual dysphoric disorder (PMDD); SD, standard deviation; IQR: interquartile range (25th percentile – 75th percentile).

^aNumbers may not add up to the total number due to missingness.

^bSelf-perceived score using the MacArthur Scale of Subjective Social Status (Galvan et al. 2023); participants can choose a ladder score from 0 (worst off) to 9 (best off) reflecting on where they think they stand at this time in their life relative to other people around them.

^cExercise minutes per week include any moderate to vigorous leisure time activity, such as brisk walking, running, cycling, dancing, strength training, playing soccer, etc.

^dPSS-4 score based on Cohen, S., Kamarck, T., & Mermelstein, R. (1983). A global measure of perceived stress. *Journal of Health and Social Behavior*, 24, 385–396. Higher scores are indicative of more stress.

Table 3.
Mean cycle length by age group, results from the linear mixed effect models with random participant-specific intercepts.

Age	Group 1: With PCOS (18,875 cycles)		Group 2: Early-life irregular (28,054 cycles)		Group 3: Early-life regular (113,277 cycles)	
	Model 1 ^a	Model 2 ^b	Model 1 ^a	Model 2 ^b	Model 1 ^a	Model 2 ^b
Under 20	36.4 (34.0, 39.1)	33.6 (29.9, 37.3)	33.8 (32.7, 34.8)	31.7 (29.7, 33.7)	30.4 (30.0, 30.8)	29.5 (28.7, 30.2)
20–24	35.7 (34.8, 36.7)	32.7 (29.8, 35.6)	32.8 (32.3, 33.3)	30.8 (29.1, 32.5)	30.3 (30.1, 30.5)	29.5 (28.8, 30.2)
25–29	34.4 (33.8, 35.1)	31.2 (28.4, 34.0)	32.2 (31.8, 32.7)	30.3 (28.6, 32.0)	30.0 (29.9, 30.2)	29.3 (28.7, 30.0)
30–34	33.4 (32.8, 33.9)	30.1 (27.4, 32.9)	31.5 (31.1, 32.0)	29.8 (28.1, 31.5)	29.5 (29.3, 29.6)	28.9 (28.2, 29.5)
35–39	32.6 (32.1, 33.2)	29.5 (26.8, 32.3)	30.7 (30.2, 31.1)	28.8 (27.2, 30.5)	28.7 (28.6, 28.9)	28.1 (27.5, 28.8)
40–44	31.5 (30.8, 32.2)	28.5 (25.8, 31.3)	30.1 (29.6, 30.6)	28.2 (26.5, 29.9)	28.4 (28.2, 28.6)	27.9 (27.2, 28.5)
45–49	30.8 (29.9, 31.7)	27.7 (24.9, 30.6)	30.5 (29.9, 31.2)	28.6 (26.9, 30.4)	29.0 (28.8, 29.2)	28.5 (27.8, 29.1)
50+	31.9 (30.4, 33.4)	28.7 (25.6, 31.7)	32.8 (31.8, 33.8)	31.0 (29.1, 33.0)	31.2 (30.9, 31.6)	30.7 (30.0, 31.5)

^aUnadjusted model. Results reported as estimated mean (95% CI) in days.

^bAdjusted for physical activity, sleep, stress, diet, smoking, alcohol, e-cigarette, and marijuana use. Details of these covariates are shown in Table 2. Results reported as estimated mean (95% CI) in days

Table 4.

Estimated within-individual standard deviations (as a reflection of cycle regularity), results from log-linear models for residual variance in the linear mixed effect models.

Age	Group 1: With PCOS (18,875 cycles)		Group 2: Early-life irregular (28,054 cycles)		Group 3: Early-life regular (113,277 cycles)	
	Model 1 ^a	Model 2 ^b	Model 1 ^a	Model 2 ^b	Model 1 ^a	Model 2 ^b
Under 20	10.5 (9.0, 11.9)	10.4 (8.6, 12.2)	10.6 (9.2, 11.9)	11.1 (9.3, 12.8)	6.1 (5.9, 6.3)	5.9 (5.7, 6.1)
20–24	11.0 (10.3, 11.7)	11.1 (10.3, 12.0)	9.4 (8.9, 9.8)	9.5 (9.0, 10.1)	6.4 (6.3, 6.5)	6.4 (6.3, 6.5)
25–29	10.3 (9.9, 10.7)	10.3 (9.7, 10.8)	8.2 (8.0, 8.5)	8.3 (8.0, 8.6)	6.5 (6.4, 6.6)	6.5 (6.4, 6.6)
30–34	9.2 (9.0, 9.5)	9.2 (8.8, 9.5)	7.4 (7.2, 7.6)	7.4 (7.1, 7.6)	6.1 (6.0, 6.2)	6.2 (6.1, 6.3)
35–39	9.4 (9.2, 9.6)	9.3 (9.0, 9.6)	7.5 (7.4, 7.6)	7.4 (7.3, 7.6)	6.1 (6.1, 6.2)	6.1 (6.1, 6.2)
40–44	8.4 (8.2, 8.6)	8.5 (8.2, 9.4)	7.3 (7.1, 7.5)	7.4 (7.1, 7.6)	6.6 (6.5, 6.7)	6.6 (6.5, 6.7)
45–49	7.9 (7.6, 8.1)	8.0 (6.5, 8.8)	8.2 (7.9, 8.5)	8.2 (7.8, 8.5)	8.6 (8.4, 8.9)	8.7 (8.4, 8.9)
50+	10.4 (9.7, 11.1)	10.4 (9.5, 11.2)	12.0 (10.5, 13.6)	12.2 (10.4, 14.0)	11.9 (10.9, 12.9)	11.9 (10.9, 13.0)

^aUnadjusted model. Results reported as estimated mean (95% CI) in days.

^bAdjusted for physical activity, sleep, stress, diet, smoking, alcohol, e-cigarette, and marijuana use. Details of these covariates are shown in Table 2. Results reported as estimated mean (95% CI) in days.